Functional Pathway-Centric Medical Knowledge Graph with Exclusion Logic

A System-Medicine Architecture for Differential Diagnosis Support

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Abstract

We present the design and prototype implementation of a functional pathway-centric medical knowledge graph for differential-diagnosis research. Many clinical decision support systems organize knowledge around diagnoses, symptoms, or genes as primary entities. In contrast, the proposed system uses *biological functional pathways* as its central organizing principle, linking genes, proteins, laboratory values, anatomical locations, cell types, and clinical diagnoses as secondary annotations.

This architecture supports a specific form of *exclusion reasoning*: if a functional pathway is judged sufficiently intact on the basis of laboratory markers, genes required for that pathway can be deprioritized as primary disease causes in the current patient, subject to redundancy, pleiotropy, and measurement-quality constraints. The system is implemented as a SQLite-backed prototype with a PySide6 graphical interface and integrates public data sources (Reactome, Gene Ontology, UniProt, HGNC, Uberon, Cell Ontology).

We describe the formal exclusion model—in both binary and probabilistic variants—together with its assumptions, limitations, and validation requirements. The prototype is demonstrated on a hemolysis–immunodeficiency scenario involving five functional pathways, eleven genes, and seven laboratory markers. The system is positioned as a research tool for exploring pathway-centric exclusion logic, not as a validated clinical decision support system.

Keywords: knowledge graph, differential diagnosis, functional pathways, exclusion reasoning, system medicine, Reactome, Gene Ontology

1 Introduction and Motivation

Differential diagnosis in complex, multi-system diseases often fails not only because data are missing, but because heterogeneous evidence is difficult to integrate systematically. A patient with abnormal laboratory values may have hundreds of candidate genetic causes; current clinical tools often present these as ranked lists without making full use of the logical structure of *excluded* possibilities.

The central observation motivating this work is:

> *If a biological functional pathway is sufficiently supported as intact, then genes whose pathogenic effect would be expected to disrupt that pathway can be deprioritized as primary causes of dysfunction.*

This transforms positive evidence (apparently intact pathway function) into negative diagnostic constraints (lower-priority candidate genes), potentially reducing the candidate space before more detailed differential-diagnostic review begins.

Existing systems organize knowledge differently. PhenoTips (Girdea et al., 2013) and Phenomizer (Köhler et al., 2009) match patient phenotypes to disease databases via the Human Phenotype Ontology (Robinson et al., 2008). These systems excel at symptom-to-disease matching but do not exploit pathway integrity as exclusion evidence. Pathway databases such as Reactome (Fabregat et al., 2018) and KEGG (Kanehisa et al., 2023) organize around biochemical reactions but are not designed for clinical exclusion reasoning. Gene Ontology (Ashburner et al., 2000) provides fine-grained process descriptions but lacks the clinical decision framework.

The proposed system addresses this gap by treating functional pathways as *first-class entities* in a knowledge graph, with all other biomedical concepts—genes, anatomical locations, cell types, laboratory markers, and diagnoses—attached as secondary annotations. This structural choice makes pathway-status reasoning explicit and computationally tractable while preserving a biologically interpretable intermediate layer.

1.1 Contributions

1. A **data model** centering on functional pathways as the primary organizing principle for clinical reasoning, linking six entity types through semantically typed edges.
2. A **formal exclusion framework** with both binary and probabilistic variants, enabling systematic prioritization and deprioritization of candidate genes based on pathway-integrity evidence.
3. A **prototype implementation** with data ingestion from six public biological databases, an interactive graph explorer, and a diagnosis query engine.
4. A **validation framework** specifying metrics, benchmark requirements, and comparison criteria for evaluating the approach.

2 Problem Statement and Formal Hypothesis

2.1 Formal Problem Statement

Let $\mathcal{P} = \{p_1, \ldots, p_n\}$ be the set of biological functional pathways for a given patient context. Let $\mathcal{G}$ be the set of candidate genes. Let $\mathcal{E}(p_i) \subseteq \mathcal{G}$ be the set of genes *essential* to pathway p_i. Let $s(p_i) \in \{0,1\}$ be the observed pathway status: 1 = intact, 0 = disrupted or unknown.

Definition 1 (Binary Exclusion Rule)**.** *If $s(p_i) = 1$, then $\forall\, g \in \mathcal{E}(p_i)$: gene g is flagged for* exclusion *from the current primary-cause candidate set, provided g is essential only in pathways that are all assessed as intact:*

$$\textit{excluded}(g) \iff \forall\, p_i \text{ s.t. } g \in \mathcal{E}(p_i) : s(p_i) = 1 \tag{1}$$

Definition 2 (Probabilistic Exclusion Score)**.** *For a gene g, the exclusion confidence is:*

$$\textit{excl}(g) = \prod_{i:\, g \in \mathcal{E}(p_i)} c(p_i) \cdot b(g) \tag{2}$$

where $c(p_i) \in [0,1]$ is the pathway status confidence (intact: 0.95, unknown: 0.50, disrupted: 0.05), and $b(g) \in [0,1]$ is a multi-pathway bonus factor:

$$b(g) = \min\Big(1.0,\ 0.7 + 0.1 \cdot |\{p_i : g \in \mathcal{E}(p_i) \wedge s(p_i) = 1\}|\Big) \tag{3}$$

The suspicion score *incorporates gene redundancy:*

$$\textit{susp}(g) = \Big(1 - \textit{excl}(g)\Big) \cdot r(g) \tag{4}$$

where $r(g) \in \{1.0, 0.85, 0.6, 0.3\}$ maps the gene's redundancy degree (none, low, medium, high).

2.2 Core Hypothesis

> *Pathway-status-based exclusion logic systematically reduces the candidate gene set for complex rare disease cases by at least 30% relative to phenotype-only filtering, when evaluated on a curated benchmark dataset.*

This is a specific, falsifiable hypothesis. The 30% threshold is used here as a pragmatic target for a meaningful reduction in review workload; a null result ($< 10\%$ reduction) would indicate that the exclusion logic adds little value beyond existing filtering tools.

The candidate reduction rate (CRR) is defined as:

$$\text{CRR} = 1 - \frac{|\mathcal{G}_{\text{after exclusion}}|}{|\mathcal{G}_{\text{before exclusion}}|} \tag{5}$$

3 System Architecture

3.1 Data Model

The central entity is the **FunctionalPathway** node, representing a discrete biological process with observable inputs and outputs. Figure 1 illustrates the complete data model.

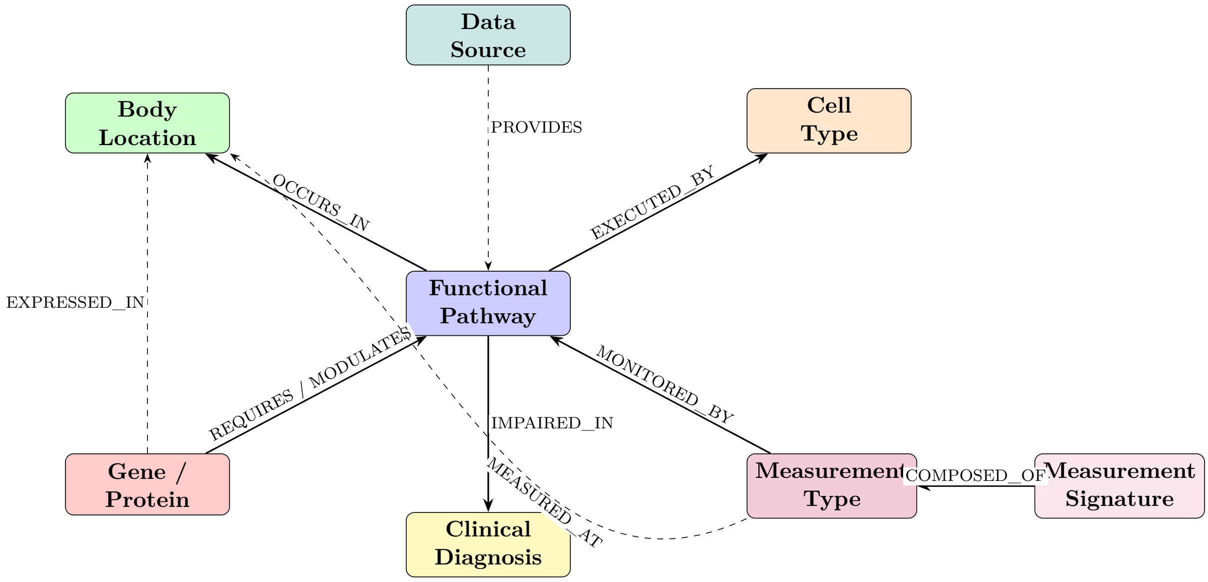

Figure 1: Entity-relationship diagram of the pathway-centric knowledge graph. Solid arrows indicate primary relationships through the pathway hub; dashed arrows indicate secondary cross-links.

Each entity type carries domain-specific attributes:

- **FunctionalPathway**: name, description, primary output, temporal class (acute or chronic), status (intact, disrupted, or unknown), and external ID (Reactome/GO).
- **BodyLocation** (Uberon): organ, subcompartment, circulation type, immune role.
- **CellType** (Cell Ontology): lineage, maturation stage, lifespan, mobility.
- **Gene/Protein** (HGNC/UniProt): symbol, function type (structural, regulatory, signaling, metabolic), redundancy degree, expression sites.
- **MeasurementType**: name, LOINC code, reference range, measurement site, sensitivity, specificity.
- **ClinicalDiagnosis**: name, ICD-10 code, description.

Edge types carry semantic labels (Table 1) and, for gene–pathway edges, an `is_essential` flag that distinguishes modulatory from essential gene contributions.

Table 1: Semantic edge types in the knowledge graph.

Source	**Target**	**Relation**
Pathway	BodyLocation	`OCCURS_IN`
Pathway	CellType	`EXECUTED_BY`
Gene	Pathway	`REQUIRES` (essential) / `MODULATES`
Measurement	Pathway	`MONITORS_OUTPUT_OF`
Measurement	BodyLocation	`REPRESENTS_ACTIVITY_IN`
Pathway	Diagnosis	`DISRUPTION_LEADS_TO`
Gene	BodyLocation	`EXPRESSED_IN`

3.2 Implementation

The prototype uses **SQLite** as the backend, using relational and association tables to emulate graph structure (Figure 2). The system comprises four main modules:

1. **Ingestion module**: Downloads and parses six public data sources (Section 3.3), with a manifest-based tracking system to avoid redundant downloads.

2. **Engine module**: Implements the exclusion logic (binary and probabilistic), graph traversal queries, pattern detection for measurement signatures, and human-readable explanation generation.

3. **GUI module**: A PySide6-based desktop application with four tabs: Graph Explorer (interactive network visualization via NetworkX), Exclusion Panel (pathway status management and analysis), Diagnosis Query (measurement-driven hypothesis generation), and Data Browser (raw table inspection).

4. **Database module**: Schema definition, CRUD helpers, and edge-linking functions with foreign key constraints.

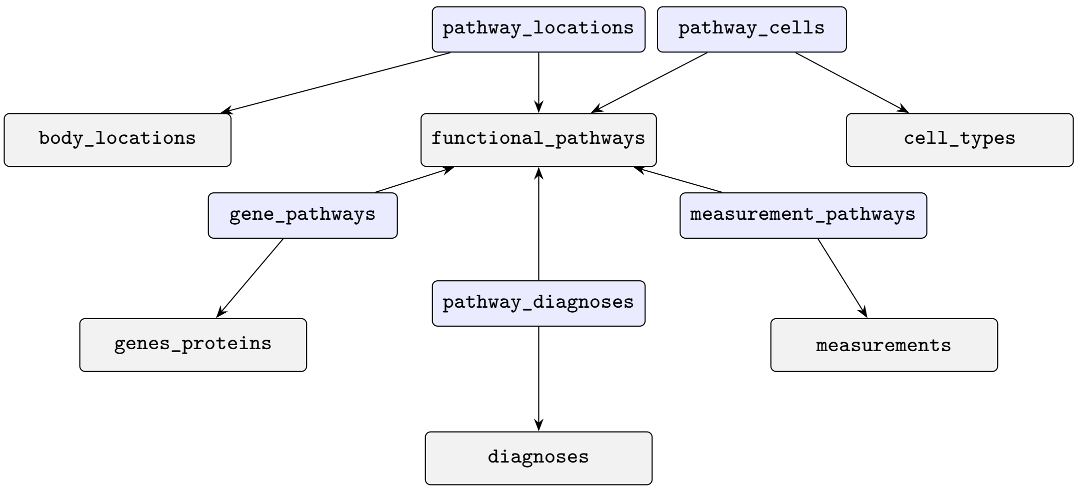

Figure 2: SQLite relational schema. Gray boxes are entity tables; blue boxes are association (edge) tables with foreign keys.

A dedicated **Graph Explorer** (NetworkX-based) visualizes subgraphs with on-demand neighborhood loading. Graph layout is computed in a background thread to maintain GUI responsiveness. Nodes are color-coded by entity type; pathway nodes additionally display their status (intact/disrupted/unknown) through border colors.

Note on graph database choice. The current SQLite implementation enables rapid prototyping but introduces known limitations for deep graph traversal (see Section 6). A production system would benefit from Neo4j or a comparable native graph database. The prototype demonstrates the conceptual model; performance comparisons with graph-native backends are planned.

3.3 Data Sources

The system integrates six public biological databases, each contributing a specific entity type to the knowledge graph:

Table 2: Integrated data sources and their roles.

Source	Entity Type	Format	Status
Reactome (Fabregat et al., 2018)	Pathways, reactions	TSV	Integrated
Gene Ontology (Ashburner et al., 2000)	Biological processes	OBO, GAF	Integrated
UniProt (The UniProt Consortium, 2023)	Proteins, gene links	TSV	Integrated
HGNC (Braschi et al., 2019)	Gene nomenclature	TSV	Integrated
Uberon (Mungall et al., 2012)	Anatomical locations	OBO	Integrated
Cell Ontology (Diehl et al., 2016)	Cell types	OBO	Integrated
KEGG (Kanehisa et al., 2023)	Pathway maps	REST API	Planned
OMIM (Amberger et al., 2019) / HPO (Robinson et al., 2008)	Disease–gene links	—	Planned

Data ingestion follows a defined order (HGNC → Uberon → Cell Ontology → UniProt → Reactome → GO) to satisfy foreign key dependencies. Each source is tracked via a manifest file recording download timestamps, SHA-256 checksums, and import status.

4 Exclusion Logic

4.1 Binary Exclusion

The binary exclusion engine (Definition 1) iterates over all genes marked as essential in at least one pathway. For each gene, it retrieves all pathways in which the gene is essential and checks whether *all* of those pathways are marked "intact." If so, the gene is excluded; otherwise, it remains a candidate.

Algorithm 1 Binary Exclusion Analysis

Require: Set of pathways $\mathcal{P}$ with status $s(p_i)$, essential gene mapping $\mathcal{E}$
Ensure: Sets $\mathcal{G}_{\text{excluded}}$, $\mathcal{G}_{\text{candidate}}$

$\mathcal{I} \leftarrow \{p_i \in \mathcal{P} : s(p_i) = \text{intact}\}$
for all $g \in \bigcup_{p \in \mathcal{P}} \mathcal{E}(p)$ **do** ▷ All essential genes
    $P_g \leftarrow \{p \in \mathcal{P} : g \in \mathcal{E}(p)\}$ ▷ Pathways requiring g
    if $P_g \subseteq \mathcal{I}$ **and** $P_g \neq \emptyset$ **then**
        $\mathcal{G}_{\text{excluded}} \leftarrow \mathcal{G}_{\text{excluded}} \cup \{g\}$
    else
        $\mathcal{G}_{\text{candidate}} \leftarrow \mathcal{G}_{\text{candidate}} \cup \{g\}$
    end if
end for

Confidence in exclusion depends on the number of intact pathways: a gene excluded via multiple independent intact pathways has stronger support than one excluded via a single pathway.

4.2 Probabilistic Exclusion

The probabilistic engine (Definition 2) extends the binary model by assigning continuous confidence scores. Three factors contribute:

1. **Pathway evidence**: Each pathway status maps to a confidence value (intact: 0.95, unknown: 0.50, disrupted: 0.05). The combined pathway confidence for a gene is the product over all its essential pathways.

2. **Multi-pathway bonus**: Genes essential in multiple intact pathways receive a bonus factor, reflecting stronger evidence from independent sources.

3. **Redundancy weighting**: The suspicion score is modulated by the gene's redundancy degree. Genes with no functional backup (redundancy = "none") receive full suspicion weight ($r = 1.0$); highly redundant genes receive reduced weight ($r = 0.3$).

Genes are categorized by exclusion confidence: high (≥ 0.8), moderate (0.3–0.8), or suspect (< 0.3). This tripartite classification is intended to support ranked review rather than to produce an automatic clinical decision.

4.3 Pattern Detection

In addition to exclusion reasoning, the system implements **measurement signature detection**. A signature is a predefined pattern of abnormal laboratory values (e.g., "hemolysis signature": low haptoglobin + high LDH + high indirect bilirubin + high reticulocytes + low hemoglobin). Each component carries a weight reflecting its diagnostic importance.

Given a set of abnormal measurements, the system computes a weighted match score against all defined signatures:

$$\text{match}(S) = \frac{\sum_{c \in S_{\text{matched}}} w_c}{\sum_{c \in S_{\text{all}}} w_c} \tag{6}$$

Signatures with match ≥ 0.3 are reported, enabling rapid identification of known clinical patterns.

5 Hemolysis and Immunodeficiency Demonstration

The prototype includes a curated seed dataset modeling the intersection of hereditary spherocytosis, autoimmune hemolytic anemia, and agammaglobulinemia. This scenario was chosen because it involves multiple interacting pathways with shared anatomical locations (spleen, bone marrow) and demonstrates the exclusion logic across independent functional domains.

5.1 Functional Pathways

Five pathways are modeled (Table 3), spanning two clinical domains (hematology and immunology) with distinct temporal characteristics.

Table 3: Functional pathways in the demonstration scenario.

Pathway	Output	Temporal
Erythrocyte membrane integrity	Stable biconcave RBCs	Chronic
Erythrocyte degradation (hemolysis)	Free Hb $\rightarrow$ bilirubin	Acute
Splenic filtration	Removal of defective cells	Acute
Early B-cell differentiation	Functional pre-B cells	Chronic
Antibody production	IgG, IgM, IgA	Both

5.2 Gene–Pathway Essentiality

Eleven genes are linked to these pathways with explicit essentiality annotations (Table 4). The essentiality flag is the critical attribute enabling exclusion logic: only essential genes can be excluded when their pathway is intact.

Table 4: Genes and their pathway essentiality in the demonstration scenario.

Gene	Pathway	Essential	Redundancy
ANK1	Membrane integrity	Yes	None
SLC4A1	Membrane integrity	Yes	None
SPTA1	Membrane integrity	Yes	Low
SPTB	Membrane integrity	Yes	None
EPB42	Membrane integrity	Yes	Low
HMOX1	Hemolysis	Yes	Low
HP	Hemolysis	No	Low
BLNK	B-cell differentiation	Yes	None
CD19	B-cell differentiation	Yes	None
PAX5	B-cell differentiation	Yes	None
NOTCH2	Splenic filtration, B-cell diff.	Yes / No	Low

5.3 Exclusion Scenario

Consider a patient presenting with:

- Low hemoglobin (anemia)
- Low haptoglobin (hemolysis marker)
- High LDH, high indirect bilirubin (hemolysis)
- Normal B-cell count (CD19+)
- Normal IgG levels

The system derives:

1. Normal B-cell count $\Rightarrow$ B-cell differentiation pathway *intact* $\Rightarrow$ **exclude** BLNK, CD19, PAX5.
2. Normal IgG $\Rightarrow$ antibody production pathway *intact*.
3. Hemolysis markers abnormal $\Rightarrow$ hemolysis pathway *disrupted*.
4. NOTCH2 is essential in splenic filtration (status unknown) and non-essential in B-cell differentiation $\Rightarrow$ **remains candidate**.
5. Membrane genes (ANK1, SLC4A1, SPTA1, SPTB, EPB42) are essential only in the membrane integrity pathway (status unknown/disrupted) $\Rightarrow$ **remain candidates**.

Result: 3 of 10 essential genes are deprioritized (CRR = 30%), matching the hypothesis threshold in this demonstration case. The remaining candidates include the membrane-structural genes associated with hereditary spherocytosis.

6 Assumptions and Limitations

6.1 Independence Assumption

The binary exclusion model assumes that pathways are *functionally independent*: an intact pathway p_i excludes its essential genes regardless of the status of other pathways. This assumption is **known to be false** in many biological contexts:

- Many genes participate in multiple pathways (*pleiotropy*). A gene excluded via pathway p_i may still be causative via a different pathway p_j that is not assessed. The model addresses this by requiring *all* pathways of a gene to be intact before exclusion.
- Compensatory mechanisms can maintain measured pathway outputs even when an essential gene is partially dysfunctional.
- The binary status $s(p_i) \in \{0, 1\}$ is a simplification; real pathway activity exists on a continuum.

The probabilistic model partially mitigates these limitations through continuous confidence scores, but the weight parameters are currently assigned heuristically.

6.2 Essentiality Definition

"Essential gene" is defined operationally as a gene annotated in Reactome or curated sources as *required* for at least one reaction in the pathway. This definition is intentionally conservative and should be refined using curated essentiality databases (e.g., DepMap, CRISPR essentiality screens) and disease-specific expert review.

6.3 Database Backend

The SQLite implementation suffices for the current prototype ($\sim$17 MB database, <2000 pathways) but introduces limitations for full-scale deployment:

- Deep graph traversals (depth > 3) require multiple JOIN operations that scale poorly.
- No native support for shortest-path or pattern-matching queries available in graph databases (e.g., Neo4j Cypher).
- Concurrent write access is limited by SQLite's file-level locking.

6.4 Data Completeness

Not all pathways have adequate Reactome coverage. Orphan pathways, tissue-specific metabolism, and recently characterized mechanisms may not be represented. The Gene Ontology integration adds breadth but at the cost of granularity—many GO biological process terms are too generic to serve as meaningful functional pathways.

6.5 EHR Interoperability

A practical deployment of exclusion-based clinical decision support requires integration with electronic health record (EHR) systems to obtain structured patient laboratory data. Without interoperable access to real-time laboratory values via standards such as HL7 FHIR (HL7 International, 2023), the system remains limited to manually entered measurements. EHR interoperability is a well-documented challenge in clinical informatics; addressing it is beyond the scope of this concept paper but constitutes a critical requirement for any future clinical application.

6.6 Measurement Redundancy and Contextuality

The probabilistic exclusion formula (Definition 2) assumes that laboratory measurements provide independent evidence about pathway status. However, Dzhafarov and Kujala's Contextuality-by-Default (CbD) framework (Dzhafarov and Kujala, 2016) warns that parallel measurements may exhibit informational redundancy rather than genuine independence. When two laboratory values decline simultaneously, this may reflect redundant measurement of the same pathway output rather than independent evidence of a shared genetic cause (pleiotropy). A rigorous treatment would require a CbD-inspired correction term in the exclusion score that accounts for measurement context overlap. This refinement is planned for future versions of the probabilistic model.

7 Related Work

7.1 Phenotype-Driven Diagnosis Support

Phenomizer (Köhler et al., 2009) and PhenoTips (Girdea et al., 2013) represent the state of the art in phenotype-driven diagnosis. Both systems match patient phenotypes (HPO terms) to disease databases and rank candidate diagnoses by semantic similarity. They are effective for symptom-to-disease matching but do not model biological pathways or exploit pathway integrity for exclusion reasoning.

7.2 Pathway Analysis in Genomics

Gene Set Enrichment Analysis (GSEA) (Subramanian et al., 2005) and related tools identify dysregulated pathways from expression data. These methods operate at the population level and require omics data that are not typically available in routine clinical settings. The proposed system instead works with individual patient laboratory values, making it closer to standard clinical workflows, although clinical applicability remains to be validated.

7.3 Graph-Based Clinical Reasoning

Knowledge graphs for clinical decision support have been explored in several contexts. Clinical knowledge graphs built from electronic health records (Rotmensch et al., 2017) capture statistical associations between diagnoses, laboratory values, and treatments. Large-scale biomedical knowledge graphs such as Hetionet (Himmelstein et al., 2017) integrate diverse data sources for drug repurposing and hypothesis generation. However, none of these systems use pathway status as a first-class constraint in exclusion reasoning.

7.4 LLM-Augmented Biomedical Reasoning

Recent work combines large language models with biomedical knowledge graphs. ESCARGOT (Matsumoto et al., 2025) integrates LLMs with a dynamic "Graph of Thoughts" architecture and biomedical knowledge graphs for multihop reasoning tasks. While ESCARGOT demonstrates effective graph-based reasoning in biomedical contexts, it operates as a general-purpose reasoning agent without explicit exclusion logic. The proposed system shares the emphasis on knowledge graph-guided reasoning but differs fundamentally in its organizing principle: functional pathways as first-class diagnostic constraints rather than general-purpose graph traversal.

HealthGenie (Gao et al., 2025) demonstrates knowledge-driven LLM integration for personalized health guidance. Its approach organizes knowledge graphs around user profiles and dietary recommendations—a symptom-centric (or need-centric) organization. In contrast, the proposed system organizes around biological functional pathways, enabling mechanistic exclusion reasoning that is not available in symptom-centric architectures.

7.5 Comparison Summary

Table 5: Comparison of the proposed system with related approaches.

System	Primary Entity	Exclusion	Lab Values	Pathways
Phenomizer	HPO phenotypes	No	No	No
PhenoTips	Patient phenotypes	No	Limited	No
GSEA/KEGG	Pathways	No	Omics only	Yes
Hetionet	Heterogeneous	No	No	Partial
Clinical KGs	EHR data	No	Yes	No
ESCARGOT	KG + LLM	No	No	Partial
HealthGenie	KG + LLM	No	No	No
Proposed	**Pathways**	**Yes**	**Yes**	**Yes**

8 Evaluation Framework

8.1 Benchmark Dataset

A rigorous validation requires a curated benchmark of 20–50 confirmed genetic diagnoses where:

1. Patient laboratory profiles are available at the time of initial presentation (before genetic diagnosis).
2. The genetic diagnosis is confirmed (variant + phenotype + functional study).
3. The patient's pathway status can be retrospectively reconstructed from laboratory data.

Target sources include rare disease registries (EURORDIS), OMIM clinical synopsis entries, and published case series with comprehensive laboratory workups.

8.2 Evaluation-Gate Ledger

The evaluation problem is not a single accuracy score. Demonstration data, synthetic stress tests, real patient benchmarks, temporal validity, provenance, and external comparators answer different questions. Table 6 separates these layers so that a positive result in one layer is not overread as validation in another.

Table 6: Evaluation gates for pathway-certificate validation.

Gate	Minimum data object	Pass / hold rule	Overclaim prevented
Demonstration seed	Curated hemolysis–immunodeficiency subgraph with manually assigned pathway states	Expected candidate-reduction logic is reproduced and plausible disease genes are retained	No clinical performance estimate
Synthetic stress tests	Planted causal gene, simulated laboratory perturbation, and known internal ground truth	The causal gene is never excluded; non-causal genes are deprioritized only when the pathway logic warrants it	No replacement for real-case validation
Real-case benchmark	20–50 confirmed rare-disease cases with pre-diagnosis laboratory profiles	CRR ≥ 0.30 with false-exclusion rate $= 0$	No autonomous diagnostic claim
Temporal certificate gate	Age window, disease stage, observation time, and validity horizon for each pathway certificate	Hard exclusion is allowed only inside the certificate's valid time window; otherwise review is required	No timeless exclusion rule
Provenance and evidence tier	Source identifier, evidence span, database version, and confidence tier for each pathway-status claim	Each retained or excluded gene is traceable to explicit source evidence	No source-free graph inference
Comparator suite	Same case set evaluated with HPO/Phenomizer and reproducible rare-disease KG or embedding baselines	Added value over HPO-only filtering and current baseline systems is visible	No isolated benchmark claim

8.3 Primary Metric

The candidate reduction rate (CRR, Equation (5)) serves as the primary success criterion. A CRR > 0.30 supports the core hypothesis; a CRR < 0.10 would indicate insufficient diagnostic value.

Safety constraint: The confirmed causal gene must *never* be excluded (false exclusion rate $= 0$). Any false exclusion in the benchmark invalidates the model for clinical application.

8.4 Alternative Validation: Synthetic Patient Scenarios

An alternative validation pathway addresses the scarcity of annotated real-world datasets for rare diseases. Synthetic patient scenarios can be constructed by introducing targeted perturbations in the pathway graph—for example, simulating an ANK1 defect by setting the erythrocyte membrane integrity pathway to "disrupted" and deriving internally consistent laboratory values. Because the causal gene is defined *a priori*, the exclusion engine can be evaluated against exact internal ground truth: the system should deprioritize non-causal genes (measuring CRR) while retaining the planted causal gene (measuring false exclusion rate). This approach complements, but cannot replace, a real-case benchmark; it is useful for stress-testing the algorithmic logic rather than for estimating clinical performance.

8.5 Secondary Metrics

- **Rank improvement**: Position of the correct gene in the ranked candidate list after exclusion filtering.

- **Coverage**: Percentage of benchmark cases where at least one pathway can be assessed from available laboratory data.
- **Comparison**: CRR versus standard HPO-only filtering on the same case set.

9 Discussion

The pathway-centric approach addresses a structural gap in current clinical decision support: the systematic use of *negative evidence* (what is demonstrably working) to constrain the diagnostic space. This is conceptually analogous to industrial fault diagnosis, where functioning subsystems eliminate entire branches of the fault tree.

Several design decisions merit discussion:

Why pathways as the center, not genes or diagnoses? Genes are candidate causes; diagnoses are clinical endpoints. Neither alone provides the mechanistic intermediate layer needed for exclusion reasoning. Pathways represent functional units that connect genetic variation to clinical effects, making them a useful pivot for both inclusion and exclusion logic.

Binary vs. probabilistic exclusion. The binary model is conceptually simpler and easier to validate (a gene is either retained or deprioritized), but it cannot express uncertainty. The probabilistic model captures graded evidence but requires calibrated parameters. We therefore treat the binary model as the initial validation target and the probabilistic model as an exploratory extension.

Scalability. The current prototype handles the demonstration scenario ($\sim$50 nodes, $\sim$80 edges) without performance concerns. Scaling to the full Reactome database (>2400 pathways, >10,000 reactions) will require performance testing and likely a migration to a graph database backend.

10 Conclusion and Future Work

The functional pathway-centric knowledge graph represents a structurally distinct approach to differential-diagnosis research, using pathway-status evidence as explicit diagnostic constraints. The prototype implementation demonstrates technical feasibility across all core components: data ingestion from six public databases, graph modeling with semantic edge types, binary and probabilistic exclusion reasoning, pattern detection, and interactive visualization.

The core hypothesis—that pathway exclusion reduces the candidate gene space by $\geq 30\%$ without false exclusions—is testable and constitutes the primary empirical target for future work.

10.1 Key Open Tasks

1. **Benchmark dataset**: Curate 20–50 cases from rare disease registries with laboratory profiles and confirmed genetic diagnoses.

2. **Weight calibration**: Develop a data-driven method for estimating pathway confidence weights $c(p_i)$ from measurement reliability and pathway redundancy.

3. **OMIM/HPO integration**: Link disease–gene associations from OMIM and HPO annotations to enable direct comparison with phenotype-only filtering.

4. **Graph database migration**: Evaluate Neo4j or similar backends for scalability beyond the SQLite prototype.

5. **Clinical tool comparison**: Benchmark against Phenomizer on the same case set to quantify the added value of pathway-centric exclusion.

6. **Temporal modeling**: Extend the framework to capture disease progression and time-dependent pathway status changes.

Data and Software Availability

The prototype source code, including database schema, ingestion pipelines, exclusion engines, and GUI application, is available in the public GitHub repository. The system integrates exclusively public data sources; no patient data is used or required.

AI Disclosure

AI-assisted tools (Claude/Anthropic and GPT/Codex) were used as non-authorial support tools during preparation of this manuscript, particularly for literature organization, code development support, text structuring, translation assistance, consistency checks, and LaTeX/PDF quality assurance. These tools did not serve as authors, co-authors, acknowledgement recipients, ground-truth sources, or independent validation authorities. The conceptual design, scientific argumentation, formal model development, final editorial decisions, and responsibility for all content lie exclusively with the author.

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